



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

EXPENSES TRACKING SYSTEM

The domain of the Project:
JAVA BASED WEB APPLICATION (SPRING BOOT)

Team Mentors (and their designation):
CHANDHANA TAGADI
PRUTHVI RAJ

Team Members:
Mr. HARI PRASATH M

Period of the project
DECEMBER 2025 to FEBRUARY 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “**EXPENSES TRACKING SYSTEM**” has been mentored by **T.CHANDHANA & PRUTHVIRAJ**, organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms.HARI PRASATH M

Signature

Mentor's Name

CHANDHANA TAGADI

Mentor'sName

PRUTHVI RAJ

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

Executive Summary – Expense Tracking System

Introduction

In today's busy lifestyle, individuals often find it difficult to keep track of their daily spending. Small expenses can easily go unnoticed and slowly affect financial stability. This project focuses on developing an Expense Tracking System that helps individuals monitor and manage their expenses in a simple, organized, and effective way.

Objective

The main objective of this system is to help individuals clearly understand how and where their money is being spent. It aims to provide a reliable platform for tracking income and expenses, encouraging better financial planning and control over personal spending habits.

Methodology

The system is designed to collect and manage expense data through multiple methods. Users can manually enter their expenses, while the system can also read transaction details from SMS and Gmail. All collected data is securely stored and automatically processed to calculate total expenses and provide meaningful summaries through an easy-to-use interface.

Key Findings

The project shows that individuals can easily monitor their daily, weekly, and monthly expenses using this system. It reduces the burden of remembering transactions and avoids manual calculations. By regularly viewing their spending patterns, users can identify unnecessary expenses and develop better money management habits.

Benefits

This system helps users stay aware of their financial activities and supports smarter decision-making. It provides clarity on total spending, improves financial discipline, and saves time by automating expense tracking and calculations.

Recommendations

To further improve the system, features such as visual charts, budget limit alerts, and savings goal tracking can be added. Mobile accessibility can also enhance user convenience and make expense monitoring available anytime and anywhere.

Conclusion

Overall, the Expense Tracking System acts as a personal financial assistant for individuals. It simplifies expense monitoring, promotes better budgeting, and helps users take control of their finances in a practical and effective manner.



Introduction

Background and Context of the Project

Nowadays, many people find it difficult to keep track of their daily expenses because spending happens in small amounts through cash, online payments, and card transactions. Over time, this makes it hard to understand where the money goes. This project was created to provide a simple system that helps individuals record and monitor their expenses in one place and stay aware of their financial activities.

Problem Statement / Goals of the Project

The main problem is that individuals often forget to note down their expenses or calculate their total spending, which can lead to poor financial management. The goal of this project is to develop an easy-to-use Expense Tracking System that helps users record, track, and understand their income and expenses clearly. It aims to help individuals control unnecessary spending and manage their money more effectively.

Scope and Limitations of the Project

This system is designed mainly for individual use to track personal expenses and income. It can collect data through manual entry, SMS, and Gmail transactions and provide total expense summaries. However, it may not support multiple users, business-level accounting, or advanced financial planning features. It also depends on the accuracy of the available SMS and email transaction data.

Innovation Component in the Project

The innovative part of this project is the automatic collection of expense details from SMS and Gmail, which reduces the need for manual entry. This makes expense tracking faster, easier, and more accurate. By combining automatic data reading with a simple interface, the system helps users monitor their spending habits in a smarter and more convenient way.



Project Objectives

Objectives and Goals of the Project

The main objective of this project is to create a simple and user-friendly Expense Tracking System that helps individuals manage their daily finances more effectively. The system is designed to make it easy for users to record their income and expenses, understand their spending habits, and take better control of their money.

Another important goal is to reduce the effort of manual tracking by automatically collecting expense details from sources like SMS and Gmail, while still allowing manual entry when needed. The system also aims to analyse expenses without requiring users to type every detail, by reading transaction messages and identifying spending patterns automatically.

In addition, the project focuses on improving convenience by allowing users to upload or capture photos of bills/receipts so the system can extract and record expense details from them. This helps users track expenses even when they only have a receipt image.

The system is also designed to send WhatsApp alerts to users with updates about their expenses, such as total spending, unusual transactions, or budget limit warnings. Clear summaries and automatic analysis will help users make better financial decisions and avoid unnecessary spending.

Expected Outcomes and Deliverables

By the completion of this project, the main outcome is to provide a fully working and reliable Expense Tracking System that allows individuals to monitor their daily, weekly, and monthly expenses in one place. The system is expected to automatically analyse spending data collected from SMS, Gmail, and receipt images, reducing the need for manual input and making expense tracking more accurate and effortless.

Users will be able to clearly understand where their money is going through automatic calculations, expense summaries, and categorized spending insights. This will help them identify unnecessary expenses and improve their financial planning.

The final deliverables of the project include a complete application with the following features: automatic expense data collection, expense analysis without manual entry, receipt/photo-based expense capture, total expense calculation, WhatsApp alerts for spending updates, and a simple, user-friendly interface to view all financial details.

Overall, the expected result is a smart and practical personal finance tool that supports users in building better financial habits, staying aware of their spending, and managing their money in a more organized, simple, and stress-free way.



Methodology and Results

- Methods/Technology used

Methods / Technology Used

The Expense Tracking System is developed using a combination of simple and modern technologies to make expense monitoring automatic, accurate, and easy for individual users. The project follows a structured method where expense data is collected from different sources, processed in the backend, stored securely, and then displayed to the user in a clear format.

The backend of the system is built using Java and Spring Boot, which handle the main processing, business logic, and API development. These APIs are responsible for collecting expense details, analysing transactions, calculating totals, and sending the processed data to the frontend.

For data storage, PostgreSQL is used as the database to safely store income details, expense records, and calculated summaries. This ensures that all financial data is organized and can be accessed whenever required.

The system uses multiple methods to collect expense data automatically. It reads transaction details from SMS and Gmail, and it can also extract expense information from uploaded images or photos of bills and receipts. These details are processed using pattern matching and data extraction techniques, which reduce the need for manual input.

The frontend is developed using HTML, CSS, and JavaScript to create a simple and user-friendly interface where users can view their expenses, totals, and financial summaries. The frontend communicates with the backend through REST APIs to fetch updated data.

Additionally, the system supports sending WhatsApp alerts to notify users about their spending summaries, unusual transactions, or budget-related updates. Overall, the combination of these technologies helps in creating an automated, efficient, and easy-to-use expense tracking solution.

- Tools/Software used
- Data collection approach (if applicable)

The development of the Expense Tracking System used a set of reliable tools and technologies to ensure smooth design, development, and testing of the application. These tools helped in building a complete system that collects, processes, and displays expense data in an efficient way.

Programming and Development Technologies:

The core of the project was developed using Java with the Spring Boot framework, which was used to create the backend logic and REST APIs for handling expense data, calculations, and system operations.

For the frontend, HTML, CSS, and JavaScript were used to design a simple and interactive user interface where users can view their expenses, totals, and financial summaries.

**Database:**

PostgreSQL was used as the main database to store all expense records, income details, and calculated results in a secure and structured format. It helps in maintaining data safely and retrieving it whenever required.

Development Environment:

IntelliJ IDEA was used as the primary IDE for writing, testing, and managing the Spring Boot and Java code. It provided helpful features like debugging, auto-suggestions, and easy project management, which made development faster and more organized.

API Testing Tools:

Postman and Swagger were used to test and verify the REST APIs. These tools helped ensure that the frontend, built using HTML, CSS, and JavaScript, was properly communicating with the Spring Boot backend and that data was being transferred correctly.

Build and Dependency Management:

Maven was used to manage project dependencies and to build the Spring Boot application efficiently. It helped maintain all required libraries and ensured smooth project setup.

Browser Tools:

Web browsers such as Google Chrome were used to run and test the frontend interface, check API responses, and confirm proper connection between the frontend, backend, and PostgreSQL database.

Together, HTML, CSS, JavaScript, Spring Boot, and PostgreSQL formed the core technologies of the project, while tools like IntelliJ, Postman, Swagger, and Maven supported development, testing, and smooth execution of the Expense Tracking System.

Data Collection Approach

The Expense Tracking System collects data in simple and practical ways to help users monitor their financial activities with minimum effort. It combines both manual and automatic methods to ensure that all types of expenses are captured and analysed properly.

Manual Entry:

Users can directly enter their daily expenses and income through the system interface. This is especially useful for recording cash transactions or small expenses that are not reflected in SMS or email messages. Manually entered data is stored and included in the overall expense analysis.

SMS-Based Collection:

The system reads bank transaction details from SMS messages and automatically extracts important information such as amount, date, and transaction type using pattern matching techniques. This reduces the need for users to manually enter digital payment expenses.



Gmail-Based Collection:

Transaction details are also collected from bank-related emails in Gmail. The system scans emails, identifies useful expense information, and stores it automatically. This helps in tracking online purchases, bill payments, and card transactions.

Photo/Receipt-Based Collection:

Users can upload or capture photos of bills and receipts. The system analyses the image, identifies key details like amount and date, and records the expense. This feature helps track spending even when only a receipt is available.

Data Storage and Analysis:

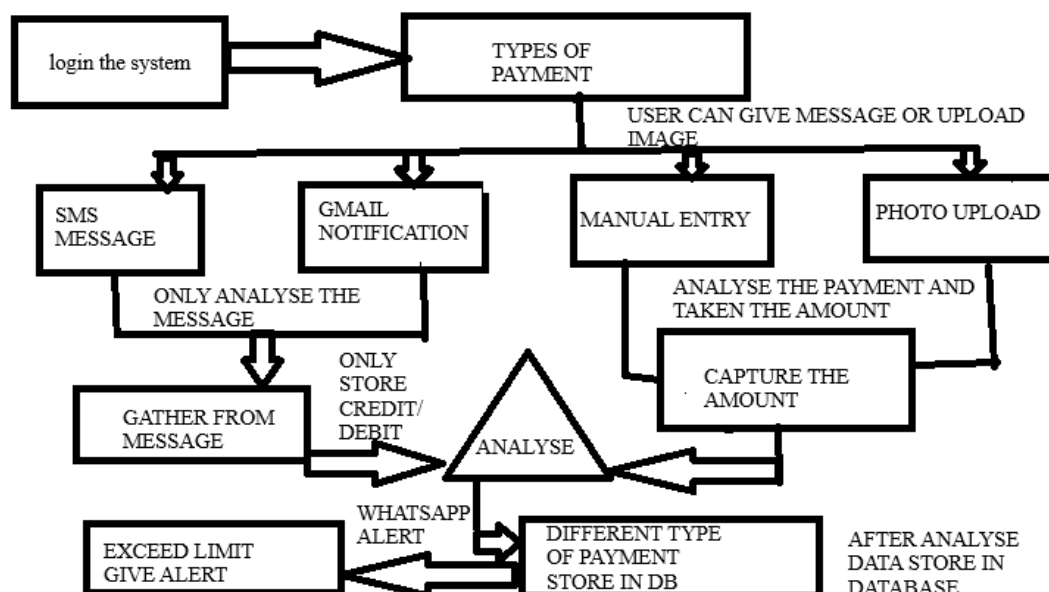
All collected data — whether entered manually, extracted from SMS, Gmail, or analysed from photos — is securely stored in the database. The system processes this information to calculate total expenses, income, and spending summaries automatically.

WhatsApp Alerts:

Based on the collected and analysed data, the system can send WhatsApp alerts to users. These alerts can include expense summaries, spending updates, or notifications when expenses increase or cross a certain limit.

This multi-source data collection approach makes the system more accurate, convenient, and user-friendly, while reducing the need for manual tracking and helping users stay aware of their spending at all times.

- **Project Architecture (Clearly describe the architecture)**





Final project working screenshots along with supporting explanation

The screenshot shows a web browser window with the URL `C:/Users/ELCOT/Documents/index.html`. The page displays a registration form titled "Expense Tracker System". The form includes input fields for "NAME:", "AGE:", "INCOME:", and "Monthly Goal:", each with a placeholder text "Enter your Name", "Enter your age", "Enter income", and "Enter monthly goal" respectively. Below the input fields are two buttons: "SUBMIT" and "VIEW PROFILE".

The screenshot shows a web browser window with the URL `C:/Users/ELCOT/Downloads/sms.html`. The page displays a dashboard titled "SMS Expense Tracking System". It features a section "Paste Bank SMS" with a text area for pasting bank SMS. Below this is an "Analyze & Save" button. The dashboard also displays four summary cards: "Total Credit ₹ 0" (green), "Total Debit ₹ 0" (red), "Net Balance ₹ 0" (blue), and "Latest Bank Balance ₹ 0" (purple). A "Refresh" button is located below these cards. At the bottom, there is a table with the following headers: "ID", "Type", "Amount", "Date", and "Available Balance".



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Gmail Expense Dashboard

Total Credit
₹ 105728

Total Debit
₹ 9682

Net Balance
₹ 96046

EMI Total
₹ 9387

Refresh Show EMI Only Show All

ID	Amount	Type	Category	Date
1	₹ 2	CREDIT	DEPOSIT	2026-02-16 07:23:39.036869
2	₹ 3000	CREDIT	DEPOSIT	2026-02-16 07:23:47.923212
3	₹ 3500	CREDIT	DEPOSIT	2026-02-16 07:23:58.676081
4	₹ 3279	DEBIT	EMI	2026-02-16 07:24:01.827027
5	₹ 6000	CREDIT	DEPOSIT	2026-02-16 07:24:04.113055
6	₹ 8	CREDIT	DEPOSIT	2026-02-16 07:24:11.174994
7	₹ 5813	DEBIT	EMI	2026-02-16 07:24:14.447049
8	₹ 40000	CREDIT	DEPOSIT	2026-02-16 07:24:18.133858

Manual Expense Dashboard

Add Expense

No file chosen

View Expense By ID

Amount:



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Photo Expense Upload

Upload Bill Photo

No file chosen

Get Amount By ID

Total Photo Expenses

Overall Expense Summary

Manual Total: ₹
Gmail Total: ₹
Photo Total: ₹
SMS Total: ₹
Grand Total: ₹
Alert Sent:
Last Updated:

- Project GitHub Link

<https://github.com/hariprasath211510/expenseTrackerSystem>



Learning and Reflection

- Document every team member's new learnings (in terms of technology, management, etc)

Team Members – New Learnings Documentation

During the development of the **Expense Tracker Web Application**, each team member gained significant technical and managerial knowledge through practical implementation and collaboration.

The **Backend Developer** learned to design and implement RESTful APIs using Spring Boot and Java. They developed CRUD operations with Spring Data JPA, connected the application to a MySQL database, and structured the project using controller, service, and repository layers. They also handled file uploads, implemented exception handling, and integrated Tesseract OCR for automated bill text extraction. This enhanced their understanding of API development, debugging, and real-time error handling.

The **Frontend Developer** gained experience in designing responsive user interfaces using HTML, CSS, and JavaScript. They integrated frontend components with backend APIs, displayed dynamic expense summaries (total expenses, income, and balance), and implemented client-side validation. This improved their understanding of UI/UX principles and API communication.

The **Database Designer** learned to design normalized database schemas with proper primary and foreign key relationships. They implemented one-to-many relationships between users and expenses, optimized queries, and improved database performance using indexing. This strengthened their knowledge of relational database management and data integrity.

The **OCR & Automation Module Developer** learned to install and configure Tesseract OCR and integrate it with the backend system. They implemented text extraction from receipts and SMS transactions and applied regular expressions to identify amounts and dates. This improved their problem-solving skills in handling inconsistent and unstructured data.

The **Team Leader / Project Manager** gained experience in project planning, task allocation, team coordination, and documentation. They ensured proper integration of all modules and managed timelines effectively. This enhanced leadership, communication, and project management skills.

Overall, the team gained hands-on experience in full-stack development, database management, API integration, automation, debugging, and collaborative software development practices.



1. HARI PRASATH – Backend Developer

Worked on backend development using Spring Boot. Designed and implemented REST APIs, handled database integration with MySQL, and managed business logic. Gained experience in API development, exception handling, debugging, and layered architecture (Controller–Service–Repository).

2. Frontend Developer

Developed responsive user interfaces using HTML, CSS, and JavaScript. Integrated frontend with backend APIs and implemented form validations. Gained experience in UI/UX design and frontend-backend integration.

3. Database & System Analyst

Designed database schema and managed data storage. Created ER diagrams and optimized queries. Improved knowledge in DBMS concepts and system architecture planning.

4. Testing & Documentation

Performed system testing and prepared project documentation. Identified bugs, validated requirements, and ensured system quality. Gained experience in software testing and technical documentation.



Conclusion and Future Scope

Recap of Objectives and Achievements

The main aim of this project was to design and develop a simple, smart, and user-friendly Expense Tracking System that helps individuals easily monitor and manage their daily financial activities. The focus was to reduce the difficulty of remembering and recording expenses, improve financial awareness, and provide a clear understanding of income and spending in one place.

This objective has been successfully achieved by building a system that supports both manual and automatic expense tracking. Users can enter expenses manually, while the system also collects transaction details automatically from sources like SMS, Gmail, and receipt images. It processes the collected data, calculates total expenses, stores the information securely, and presents it in a clear and easy-to-understand format.

With features like automatic analysis and spending summaries, the system helps users understand their spending habits and take better control of their finances in their daily life.

Overall, the project has resulted in a practical, efficient, and user-friendly solution that makes expense tracking more organized, reduces manual effort, and supports better financial management for individual users.

Future Scope of the Project

The Expense Tracking System developed in this project provides a strong foundation for personal financial management. In the future, the system can be enhanced by adding more advanced and user-friendly features to make it even more useful and intelligent.

One of the major improvements can be the development of a mobile application so that users can track their expenses anytime and anywhere. This will make the system more accessible and convenient for daily use.

The project can also be extended by adding smart visual reports such as graphs and charts to help users clearly understand their spending patterns. Features like monthly budget setting, savings goal tracking, and alerts when spending crosses a limit can support better financial planning.

Further enhancements can include improved photo analysis to automatically read bill details more accurately, better categorization of expenses, and more detailed WhatsApp notifications about daily or monthly spending summaries.



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The system can also be expanded to support multiple users, family expense tracking, and shared expense management. In the long term, advanced technologies like AI-based analysis can be added to predict future expenses and provide suggestions to save money.

With these improvements, the project has the potential to grow into a complete personal finance management solution that helps users manage their money more smartly, easily, and effectively.